



1. Industry Adoption Statistics

Docker. (2025, July 11). *2025 Docker state of app dev: Key insights revealed*. Docker Blog. <https://www.docker.com/blog/2025-docker-state-of-app-dev/>

Docker's 2025 State of Application Development report reveals that container usage among IT professionals has surged to 92%, up from 80% in 2024, marking the largest single-year adoption increase of any surveyed technology. The report further indicates that 68% of IT/SaaS companies utilize microservice-based architectures compared to only 31% in traditional industries, directly correlating higher container adoption with cloud-native development practices.

BEAM Schema: Exhibit (provides empirical data and statistics)

How this helps my proposal: These statistics demonstrate overwhelming industry adoption of containerization, providing compelling evidence that Docker has become a fundamental skill rather than an optional one. The 92% adoption rate among IT professionals strengthens my argument that UNT's CS curriculum must prepare students for industry-standard development environments. I can cite specific statistics like "container usage soared to 92% — up from 80% in our 2024 survey" to illustrate the urgency of curriculum modernization.

2. Educational Implementation

Sorrel, H. (2019, November 3). *Docker in the classroom (part 1)*. Medium. <https://sorrelharriet.medium.com/docker-in-the-classroom-part-1-3d86fea9160c>

This article provides practical insights into implementing Docker in educational settings, emphasizing that Docker is cross-platform and eliminates the need for complex IT infrastructure changes. The author demonstrates how Docker simplifies teaching database and web server concepts while maintaining standardized environments across diverse student operating systems, reducing technical support overhead for instructors.

BEAM Schema: Method (demonstrates how to implement Docker in education)

How this helps my proposal: This source offers concrete evidence that Docker integration is pedagogically feasible and actually simplifies course management rather than complicating it.

The author's statement that "Docker really isn't hard" for basic educational use cases directly addresses potential faculty concerns about implementation difficulty. I can reference this to show that existing educators have successfully integrated Docker without requiring additional courses or significant curriculum restructuring, supporting my recommendation to embed Docker within existing upper-division courses.

3. Skills Gap Research

Sánchez-Cifo, M. Á., Bermejo, P., & Navarro, E. (2024). Introducing computer science undergraduate students to DevOps technologies from software engineering fundamentals. In *Proceedings of the 46th International Conference on Software Engineering: Software Engineering Education and Training* (pp. 1-2). Association for Computing Machinery. <https://doi.org/10.1145/3639474.3640071>

This peer-reviewed research investigates the challenge of introducing DevOps practices, including CI/CD and containerization, within Computer Science fundamentals courses. The study demonstrates that demand for software engineers with DevOps skills continues to increase, yet CS majors face difficulties connecting abstract academic concepts to industry-required practical skills. Student outcomes from early DevOps exposure were "extremely positive," providing insights into bridging the academia-industry gap.

BEAM Schema: Argument (presents research-based position on curriculum needs)

How this helps my proposal: This academic research validates my central argument that a skills gap exists between CS education and industry expectations. The finding that students respond positively to early DevOps exposure supports my recommendation to integrate Docker in upper-division courses rather than waiting until capstone projects. I can cite this peer-reviewed source to demonstrate that my proposal aligns with established educational research and best practices in CS curriculum development.

4. Job Market Analysis

Kube Careers. (2024). *The state of Kubernetes jobs in 2024 Q3*. <https://kube.careers/state-of-kubernetes-jobs-2024-q3>

This comprehensive analysis of 8,772 Kubernetes-related job postings reveals that Docker is mentioned in over 60% of position requirements, making it the most frequently cited technology. The average salary for Kubernetes-skilled professionals reached \$158,134 globally, with North American positions averaging \$169,627. Software Engineer positions comprised 42% of all Kubernetes-related roles, followed by DevOps Engineer (11%) and Platform Engineer (8%) positions.

BEAM Schema: Exhibit (provides labor market data and trends)

How this helps my proposal: These job market statistics provide quantitative evidence of employer expectations and the financial benefits for graduates with containerization skills. The finding that Docker appears in 60% of Kubernetes job postings demonstrates it's a prerequisite skill, not just a "nice-to-have" capability. I can use salary data to illustrate the career advantages for UNT graduates who acquire Docker experience during their studies, strengthening the value proposition of my curriculum proposal.

5. Container Market Growth

Grand View Research. (2024). *Application container market size | Industry report, 2030*.
<https://www.grandviewresearch.com/industry-analysis/application-container-market>

The global application container market was valued at \$5.85 billion in 2024 and is projected to reach \$31.5 billion by 2030, representing a compound annual growth rate of 33.5%. This growth is driven by increasing adoption of DevOps methodologies, cloud-native technologies, and microservices architectures. The report indicates that 83% of developers participated in DevOps-related activities in Q1 2024, marking a 6% increase from 2022.

BEAM Schema: Background (establishes market context and trends)

How this helps my proposal: This market research provides broader economic context showing that containerization isn't a passing trend but a sustained industry shift with significant financial backing. The 33.5% CAGR demonstrates that Docker skills will remain relevant throughout students' careers, not just at graduation. I can reference these projections to assure stakeholders that curriculum investments in containerization will provide long-term value and won't become outdated, addressing potential concerns about technological obsolescence.